

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \ln e = 1$$

$$\lim_{x \rightarrow -\infty} e^x = 0, \quad \lim_{x \rightarrow +\infty} e^x = \infty$$

$$\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad (\text{NR method})$$

$$\ln 1 = 0, \ln \infty = \infty$$

$$\tan 0 = 0$$

$$\tan 90 = \infty$$

$$\ln 0 = -\infty$$

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0$$

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \operatorname{cosec}^{-1} x = \frac{-1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$\frac{d}{dx} a^x = a^x \ln a$$

$$\frac{d}{dx} \ln(x) = \frac{1}{x} \frac{d}{dx} x$$

$$\frac{d}{dx} x^n = x^n (1 + \ln x)$$

CARTESIAN

$$K = \frac{\frac{d^2 y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}} = \frac{y''}{\left[1 + (y')^2\right]^{3/2}}$$

$$P = \frac{1}{K} = \frac{\left[1 + (y')^2\right]^{3/2}}{y''}$$

$$K = \frac{r^2 + 2r'^2 - rr''}{(r^2 + r'^2)^{3/2}}$$

$$= \frac{r^2 + 2\left(\frac{dr}{ds}\right)^2 - r\left(\frac{d^2 r}{ds^2}\right)}{\left[r^2 + \left(\frac{dr}{ds}\right)^2\right]^{3/2}}$$

$$P = \frac{(r^2 + r'^2)^{3/2}}{r^2 + 2r'^2 - rr''}$$

Maclaurin Series

$$f(x) \approx f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n$$

Taylor Series

$$f(x) \approx f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \dots$$

Partial Derivatives Notations

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}, \quad r = \frac{\partial^2 z}{\partial x^2}$$

$$s = \frac{\partial^2 z}{\partial x \partial y}, \quad t = \frac{\partial^2 z}{\partial y^2}$$

$$\frac{\partial z}{\partial x} \approx \frac{\partial f}{\partial x} = f_x(x, y) = D_x f$$

Fundamental Trigonometry

$$\cos^2 x + \sin^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \operatorname{cosec}^2 x$$

Total Differential Coefficient: $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$ For $z = f(x, y)$

For $z = f(a, b, c)$ $dz = \frac{\partial z}{\partial a} da + \frac{\partial z}{\partial b} db + \frac{\partial z}{\partial c} dc$

- du : absolute / actual error in u
- $\frac{du}{u}$: relative error in u
- $\frac{du}{u} \times 100$: Percentage error in u

REDUCTION FORMULAE:

$$\textcircled{1} \int \sin^n x \, dx = -\frac{\cos x \sin^{n-1} x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$\textcircled{2} \int \cos^n x \, dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$\textcircled{3} \int_0^{n/2} \sin^n x \, dx = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \dots \frac{2}{3} & ; n \text{ is odd} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \dots \frac{1}{2} \cdot \frac{\pi}{2} & ; n \text{ is even} \end{cases}$$

$$\textcircled{4} \int_0^{n/2} \cos^n x \, dx = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \dots \frac{2}{3} & ; n \text{ is odd} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \dots \frac{1}{2} \cdot \frac{\pi}{2} & ; n \text{ is even} \end{cases}$$

$$\textcircled{5} \int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx$$

$$\textcircled{6} \int \cot^n x \, dx = -\frac{\cot^{n-1} x}{n-1} - \int \cot^{n-2} x \, dx$$

$$\textcircled{7} \int \sec^n x \, dx = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$$

$$\textcircled{8} \int \csc^n x \, dx = -\frac{\cot x \csc^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \csc^{n-2} x \, dx$$

$$\textcircled{9} \int \sin^p x \cos^q x \, dx = -\frac{\cos^{q+1} x \sin^{p-1} x}{p+q} + \frac{p-1}{p+q} \int \sin^{p-2} x \cos^q x \, dx$$

$\textcircled{10}$ If p is even & q is even

$$\int_0^{n/2} \sin^p x \cos^q x \, dx = \frac{(p-1)(p-3)(p-5) \dots (q-1)(q-3) \dots \pi}{(p+q)(p+q-2)(p+q-4) \dots 2}$$

For $z = f(x, y)$
 $\int \sin^p x \cos^q x \, dx = \frac{(p-1)(p-3)(p-5) \dots (q-1)(q-3) \dots \pi}{(p+q)(p+q-2)(p+q-4) \dots 2}$
 p odd & q odd
 p odd & q even
 p even & q odd

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c \quad \text{or} \quad \int (ax+b)^n dx = \frac{1}{a} \frac{(ax+b)^{n+1}}{n+1} + c$$

$$\int \frac{1}{x} dx = \ln x + c \quad \text{or} \quad \int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + c$$

$$\int \sin x dx = -\cos x + c$$

$$\int \frac{1}{\sqrt{x^2-a^2}} = \frac{1}{a} \sec^{-1} \frac{x}{a}$$

$$\int \cos x dx = \sin x + c$$

$$\int \frac{1}{\sqrt{x^2+a^2}} dx = \ln(x + \sqrt{a^2+x^2})$$

$$\int \sec^2 x dx = \tan x + c$$

$$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$\int \frac{1}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a}$$

$$\int \tan x \sec x dx = \sec x + c$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$$

$$\int f'(x) f^{-1}(x) dx = \ln f(x) + c$$

$$\int \cot x dx = \ln(\sin x) + c$$

$$\int \sec x dx = \ln|\sec x + \tan x| + c$$

Definite Integrals

$$\int \operatorname{cosec} x dx = \ln|\operatorname{cosec} x - \cot x| + c$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int \tan x dx = \ln|\sec x| + c$$

$$\text{or } -\ln|\cos x| + c$$

$$\int_a^b f(x) = \int_a^c f(x) + \int_c^b f(x) dx$$

Trigonometric Substitutions

$$\sqrt{a^2-x^2} \quad x = a \sin \theta$$

$$\sqrt{a^2+x^2} \quad x = a \tan \theta$$

$$\sqrt{x^2-a^2} \quad x = a \sec \theta$$

Integration by Parts

$$\int u dv = uv - \int v du$$

$$\int a^x dx = \frac{1}{\ln a} a^x + c$$

$$\int \sin \theta d\theta = -\cos \theta + c$$

$$\int e^{ax} [af(x) + f'(x)] dx = e^{ax} f(x) + c$$

Surface Area of rectangular box = $2(lw + lh + wh)$

• Gamma Function:

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, n > 0$$

• Beta Function $\beta(m, n)$

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

m, n are +ve, $m, n > 0$

• Relation b/w α & β

$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

$$\Gamma(1) = 1$$

$$\beta(m, m) = \int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$\Gamma(n+1) = n \Gamma(n)$$

$$\Gamma(n+1) = n!$$

$$\Gamma(n) = (n-1) \Gamma(n-1)$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\int_0^{\pi/2} \sin^n x = \frac{\sqrt{\frac{n+1}{2}} \sqrt{\pi}}{\sqrt{\frac{n+2}{2}}}$$

$$\int_0^{\pi/2} \cos^n x dx = \frac{\sqrt{\frac{n+1}{2}} \sqrt{\pi}}{2 \sqrt{\frac{n+2}{2}}}$$

$$\sec^2 z = 1 + \tan^2 z \quad \sin(-z) = -\sin z$$

$$\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$$

$$\tan(z_1 + z_2) = \frac{\tan z_1 + \tan z_2}{1 - \tan z_1 \tan z_2}$$

$$\sin z = \sin \bar{z}$$

$$\cos z = \cos \bar{z}$$

$$\int_0^{\pi/2} \sin^p x \cos^q x = \frac{\sqrt{\frac{p+1}{2}} \sqrt{\frac{q+1}{2}}}{2 \sqrt{\frac{p+q+2}{2}}}$$

$$r = \text{modulus} = |z| = \sqrt{a^2 + b^2}$$

$$\theta = \text{argument} = \tan^{-1}\left(\frac{b}{a}\right)$$

$$= \text{Arg}(z)$$

Range of complex number = principle argument, $-\pi < \theta < \pi$

$$a = r \cos \theta, b = r \sin \theta$$

$$Z = x + yi$$

$$r \cos \theta + yi \sin \theta$$

$$r(\cos \theta + i \sin \theta)$$

$$r(\cos \theta)$$

DE MOUVIER'S THEOREM

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$\Rightarrow x = \cos \theta + i \sin \theta$$

$$\Rightarrow x^n = (\cos \theta + i \sin \theta)^n$$

$$\Rightarrow \frac{1}{x^n} = \cos n\theta - i \sin n\theta$$

$$\Rightarrow x^n + \frac{1}{x^n} = 2 \cos n\theta$$

$$\Rightarrow x^n - \frac{1}{x^n} = 2i \sin n\theta$$

EXPONENTIAL FUNCTION

$$e^z = e^{x+yi} = e^x \cdot e^{yi}$$

$$= e^x (\cos y + i \sin y)$$

$$\therefore e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-iy} = \cos y - i \sin y$$

$$e^{iy} = \cos y + i \sin y$$

$$\cos y = \frac{e^{iy} + e^{-iy}}{2}$$

$$\sin y = \frac{e^{iy} - e^{-iy}}{2i}$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\tan z = \frac{e^{iz} - e^{-iz}}{i(e^{iz} + e^{-iz})}$$

HYPERBOLIC FUNCTION

$$\sinh z = \frac{e^z - e^{-z}}{2}$$

$$\tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\cosh z = \frac{e^z + e^{-z}}{2}$$

$$\cosh^2 z - \sinh^2 z = 1$$

$$\operatorname{sech}^2 z = 1 - \tanh^2 z$$

$$\operatorname{cosech}^2 z = \coth^2 z - 1$$

$$\sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2$$

$$\tanh(z_1 + z_2) = \frac{\tanh z_1 + \tanh z_2}{1 + \tanh z_1 \tanh z_2}$$

FROM CIRCULAR TO HYPERBOLIC

$$\sin(iz) = i \sinh z$$

$$\cos(iz) = \cosh z$$

$$\tan(iz) = i \tanh z$$

$$\cot(iz) = -i \coth z$$

$$\operatorname{cosec}(iz) = -i \operatorname{cosech} z$$

$$\sec(iz) = \operatorname{sech} z$$

FROM HYPERBOLIC TO CIRCULAR

$$\sinh(iz) = i \sin z$$

$$\cosh(iz) = \cos z$$

$$\tanh(iz) = i \tan z$$

$$\coth(iz) = -i \cot z$$

$$\operatorname{cosech}(iz) = -i \operatorname{cosec} z$$

$$\operatorname{sech}(iz) = \sec z$$